Tugas Fungsi Kalkulus

1. $f(x) = 1 - x^2$

a. $f(1) = 1 - x^2$
$= 1 - 1$
$= 0$

b. $f(-2) = 1 - (-2)^2$
$= 1 - 4$
$= -3$

c. $f(0) = 1 - (0)^2$
$= 1 - 0$
$= 1$

d. $f(k) = 1 - (k)^2$
$= 1 - k^2$

e. $f(-5) = 1 - (-5)^2$
$= 1 - 25$
$= -24$

f. $f(1/4) = 1 - (1/4)^2$
$= \frac{16}{16} - \frac{1}{16} = \frac{15}{16}$

g. $f(3t) = 1 - (3t)^2$
$= 1 - 3t^2$
$= 1 - 9t^2$

h. $f(2x) = 1 - (2x)^2$
$= 1 - 4x^2$

i. $f(1/t) = 1 - (1/t)^2$
$= 1 - (1/t)^2$
$= 1 - 1/t^2$

2.) $f(x) = x+3 \quad g(x) = x^2$

a. $(f+g)(2)$
$= 2+3+2^2$
$= 5+4$
$= 9$

b. $(f.g)(0)$
$= (0+3).0^2$
$= 0$

c. $(g/f)(3)$
$= \frac{3^2}{3+3} = \frac{9}{6} = \frac{3}{2}$

d. $(f \circ g)(1)$
$= f(g(1))$
$= f(1) = (x)^2 + 3 = 1+3 = 4$

e. $(g \circ f)(1)$
$= g(f(1))$
$= g(1) = (x+3)^2$
$= (1+3)^2$
$= (4)^2$
$= 16$

f. $(g \circ f)(-8)$
$= g(f(-8))$
$= g(-8) = (x+3)^2$
$= (-8+3)^2$
$= (-5)^2$
$= 25$

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3.) $f(x) = x^2+x \qquad g(x) = \frac{2}{x+3}$

a. $(f-g)(2)$

$= 2^2+2-\frac{2}{2+3}$

$= 6-\frac{2}{5} = \cancel{6} = \frac{30}{5}-\frac{2}{5} = \frac{28}{5}$

b. $(f/g)(1)$

$= \frac{1^2+1}{\frac{2}{1+3}} = \frac{2}{\frac{2}{4}} = \cancel{2}\cdot\frac{4}{\cancel{2}} = 4$

c. $g^2(3) = \left(\frac{2}{3+3}\right)^2 = \left(\frac{2}{6}\right)^2 = \frac{4}{36} = \frac{1}{9}$

d. $(f\circ g)(1)$

$= f(g(1))$

$= f(1) = \left(\frac{2}{x+3}\right)^2 + \frac{2}{x+3}$

$= \left(\frac{2}{1+3}\right)^2 + \frac{2}{1+3}$

$= \left(\frac{2}{4}\right)^2 + \frac{2}{4}$

$= \frac{4}{8} + \frac{2}{4} = \frac{4+4}{8} = \frac{8}{8} = 1$

c. $(g \circ f)(1)$

$= g(f(1))$

$= g(1) = \frac{2}{1^2+1+3} = \frac{2}{5}$

f. $\cancel{(g \circ g)(3)}$

$= \cancel{g(g(3))}$

$= \cancel{g(3)} = \cancel{\frac{2}{\frac{2}{3^2+3}+3}} = \cancel{\frac{2}{\frac{2}{12}+3}} = \cancel{\frac{2}{\frac{2}{12}}} = 2 \cdot \cancel{\frac{12}{5}} = \cancel{\frac{24}{5}} = \cancel{\frac{4}{5}}$

f. $(g \circ g)(3)$

$= g(g(3))$

$= g(3) = \frac{2}{\frac{2}{3+3}+3} = \frac{2}{\frac{2}{6}+3} = \frac{2}{\frac{1}{3}+3} = \frac{2}{\frac{1}{3}+\frac{9}{3}} = \frac{2}{\frac{10}{3}}$

$= \cancel{2}\,\frac{3}{\cancel{10}} = \frac{3}{5}$